Hwang 2022 PDAC — analysis summary

RUN 2 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Samples mixed (iLISI 9.3/38 vs 1.0 raw; falls to 6.7 at θ=20) on one continuous connected manifold; density lobes are poles of one axis, not batch islands nor a cycling compartment.

Gate 2 — module geography → GO (non-branching)

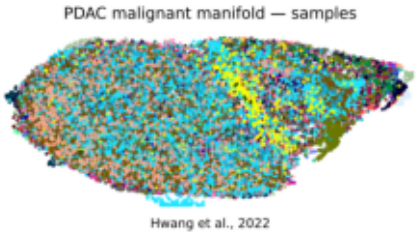
Organized single-axis geography: Gast high at root, opposes pSom/Crest spatially (Gast↔pSom r=−0.33); Crest an inverse/terminal gradient. Non-branching case → GO.

Gate 3 — hub cross-repression → NA

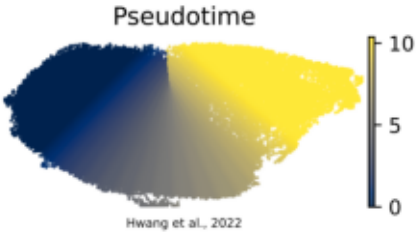
No bifurcation (scFates curve, not tree) → cross-repression scatter N/A.

STEP FIGURES

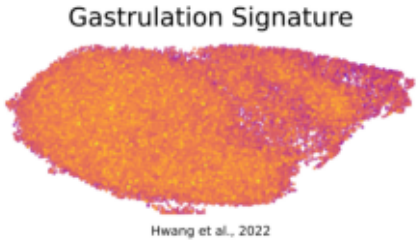
Manifold — samples



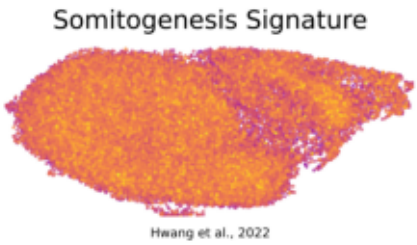
Pseudotime



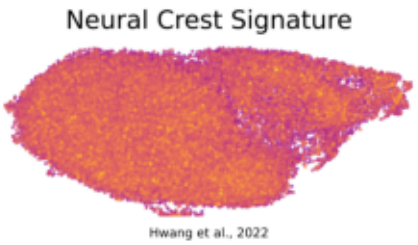
Gastrulation



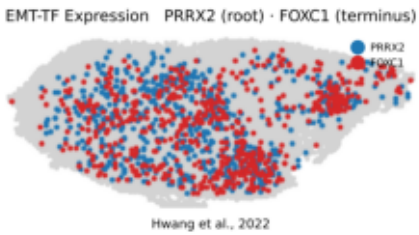
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pancreatic ductal adenocarcinoma (Hwang et al., 2022), snRNA-seq / 10x — 64,401 malignant cells across 38 tumours after threshold + island removal.

Verdict.
Pattern reproduces as the NON-BRANCHING case: one continuous gastrulation → somitogenesis / neural-crest axis, rooted at the Gast/pSom-high developmental pole. No competing arms and no cross-repression bifurcation. Geometry = Linear-pSom/Crest. Gate 1 GO / Gate 2 GO / Gate 3 NA. Module means Gast −0.005 / pSom −0.002 / Crest +0.003 (weak, spatially organized). EMT-TFs PRRX2 (root) · FOXC1 (terminus); ZEB1 most expressed (~16%).

Parameter choices

Malignant cells	64,538 raw → 64,507 (thresh) → 64,401 (islands)	**Min-cell threshold**	14 (gap-finding; smooth tail, only 24→9 gap); 5 tiny samples dropped
Integration	Harmony θ=10, n_pcs=20, spread=1.3	**Overdispersed genes**	1,664 (of 19,457 retained)
Geometry	Linear-pSom/Crest (curve: 2 tips, 0 forks, fold-free 2.72)	**Root**	Gast/pSom-high tip (cycling flat, ρ=−0.03)
Module means	Gast −0.005 / pSom −0.002 / Crest +0.003	**EMT-TFs**	PRRX2 (root) · FOXC1 (terminus); ZEB1 most expressed (~16%)